THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET A

Chapters 5–13: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity, Our Environment** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The functional unit of the kidney is: 1

(a) neuron **(b)** alveolus **(c)** nephron **(d)** villus

2. For the opening and closing of stomata, Student A says guard cells control it; Student B says xylem controls it. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

3. The breakdown of pyruvate into lactic acid occurs in: 1

(a) mitochondria with oxygen **(b)** yeast

(c) chloroplasts **(d)** muscle cells during lack of oxygen

4. The information acquired at the end of the dendrite sets off a chemical reaction that creates: 1

(a) a hormone **(b)** a synapse

(c) an electrical impulse **(d)** a reflex

5. The plant hormone that promotes cell division is: 1

(a) ethylene **(b)** auxin **(c)** abscisic acid **(d)** cytokinin

6. A student states: “Reflex actions are controlled by the spinal cord.” The correct response is: 1

(a) Correct

(b) Incorrect — they are controlled by the cerebrum

(c) Only in plants

(d) Cannot be determined

7. Asexual reproduction by budding occurs in: 1

(a) Amoeba **(b)** Plasmodium **(c)** Hydra **(d)** Planaria

8. The structure in which nutrition of the embryo occurs from the mother is: 1

(a) cervix **(b)** placenta **(c)** ovary **(d)** fallopian tube

9. Genes are segments of: 1

(a) RNA only **(b)** DNA **(c)** proteins **(d)** lipids

10. The chance of having a male child is: 1

(a) 25% **(b)** 50% **(c)** 75% **(d)** 100%

11. Which of these constitutes a food chain? 1

(a) goat, cow and elephant **(b)** grass, wheat and mango

(c) grass, goat and human **(d)** wolf, lion and tiger

12. Green plants in a terrestrial ecosystem capture about ___ of the sunlight falling on leaves. 1

(a) 10% **(b)** 50% **(c)** 1% **(d)** 90%

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. A student says that the right side of the heart carries oxygenated blood. Justify or refute. 2

14. What is a synapse? How is an impulse transmitted across it? 2

15. A student says that self-pollination and cross-pollination are the same. Justify or refute. 2

16. What is heredity? 2

17. A student says that the number of trophic levels in a food chain is usually unlimited. Justify or refute. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. How is water transported in plants? Explain the role of transpiration. 3

19. Explain spore formation in Rhizopus with a diagram (describe). 3

20. A plant with violet flowers (VV) is crossed with a white-flowered plant (vv). What will be the F1 and F2 results? Show the cross. 3

21. Why do producers always occupy the first trophic level? 3

22. Explain how traits are inherited independently, with a dihybrid example. 3

OR

Explain how sex is determined in human beings with a flow chart. 3

23. Differentiate between biodegradable and non-biodegradable substances with examples. 3

OR

Why are food chains generally short? Explain. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Draw a labelled diagram (describe) of the human brain. 5
(ii) Explain the functions of the fore-brain, mid-brain and hind-brain.

OR

(i) Describe the human nervous system and its divisions. 5
(ii) What are the functions of the spinal cord?
(iii) What is the role of the peripheral nervous system?

25. (i) Explain the 10% law of energy flow with a diagram. 5
(ii) If 10 000 J is available at producers, how much energy reaches the tertiary consumer?
(iii) Why are there fewer animals at higher trophic levels?

OR

(i) What is the ozone layer? 5
(ii) How is it being depleted?
(iii) What steps were taken internationally?
(iv) Effects of UV on humans?

END OF QUESTION PAPER Total: 50 Marks